



MGM's College of Engineering, Nanded

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Secure Trade Finance Explorer for Transaction Transparency and Compliance Insights

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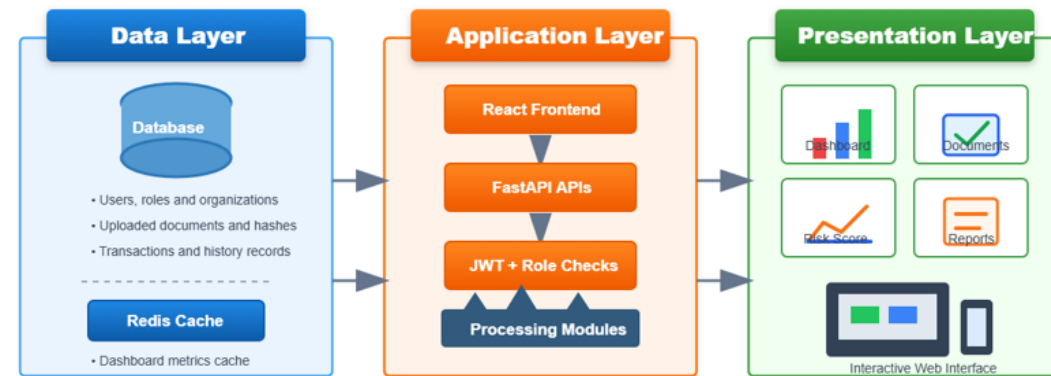
Introduction: Secure Trade Finance Management System is a web-based application developed using React and FastAPI to manage trade finance documents, transactions, users, alerts, ledgers, risk scores, and reports. The system provides secure authentication, role-based access control, document hashing, transaction tracking, risk analysis, and analytics dashboards. It helps users manage trade finance activities in a structured and transparent way. The system also maintains audit tracking, compliance records, and transaction history to improve transparency and monitoring.

System Architecture

- **Data Source Layer:**
Stores users, roles, trade documents, document hashes, transactions, alerts, ledger records, audit logs, and risk scores in the database.
- **Backend Layer:**
Handles APIs, authentication, transactions, alerts, reports, and database communication using FastAPI.
- **Processing Layer:**
Processes documents, verifies SHA-256 hashes, manages transactions, and calculates risk scores.
- **Application Layer:**
Connects frontend and backend for login, document upload, transactions, alerts, and reports.
- **Presentation Layer:**
Displays dashboards, documents, transactions, alerts, ledgers, reports, and user pages using React.

Methodology

The system authenticates users using JWT tokens and applies role-based access control. Users can upload trade documents, and the system generates SHA-256 hash values to verify document integrity. Trade transactions are created between buyers and sellers and tracked through valid status changes. The system records transaction history, creates alerts, and displays analytics and reports through the dashboard.



Conclusion: The Secure Trade Finance Management System provides a secure and organized platform for managing trade finance documents, transactions, users, ledgers, alerts, risk scores, and reports. By using document hashing, role-based access, transaction tracking, risk analysis, and analytics dashboards, the system improves transparency, monitoring, and record management in trade finance operations.

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